

CHENG-HSUAN LEE

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Robotics-focused Mechanical Engineering undergraduate | Legged robotics, robot learning, actuator design, sim-to-real testing

PROFILE

Mechanical Engineering undergraduate at National Taiwan University focused on **complete physical robot systems**. My current work in **NTUME ASR Lab** connects quadruped actuator hardware, motor communication, Isaac Lab robot-learning workflows, ROS2 deployment, and **sim-to-real debugging**. I am especially interested in legged robots whose hardware, control policies, and experimental behavior must be designed as one coupled system.

EDUCATION

National Taiwan University - B.S. in Mechanical Engineering

Taipei, Taiwan | Expected 2027

Relevant coursework: Automatic Control; Microprocessor Controlled Systems; Applied Electronics; Mechanism; Machine Design Theory; Engineering Graphics; Practice of Mechanical Engineering; Manufacturing Processes; Workshop Practice; Mechanical Engineering Laboratory I/II; Engineering Materials; Strength of Materials; Dynamics; Numerical Analysis; Fluid Mechanics; Thermodynamics; Heat Transfer.

Taipei Municipal Chien Kuo High School

Taipei, Taiwan | 2020-2023

Built early robotics foundation through CKHS Robotics Team, **FRC programming leadership**, strategy, scouting, and international competition work.

RESEARCH AND LABORATORY EXPERIENCE

NTUME ASR Lab - Learning-Based Quadruped Robot Integration

Control Systems Laboratory Experience | Rifle 12-DoF quadruped platform | Current laboratory work

Working at the boundary between **legged robot hardware**, **embedded motor-control interfaces**, and **learning-based locomotion**. My lab work focuses on making learned policies usable on imperfect physical hardware by connecting actuator/reducer design, command/readback reliability, Isaac Lab training behavior, ROS2 deployment logs, and failure analysis into one experimental loop.

- **Hardware-control integration:** contributed to a quadruped development workflow linking mechanical architecture, leg-module packaging, reducer concepts, wiring, motor communication, Isaac Lab simulation, and physical robot validation.
- **Actuator and reducer development:** designed and fabricated motor reduction gearbox concepts for leg actuation, including cycloidal and planetary reducers, while evaluating torque transmission, backlash, bearing support, manufacturability, assembly tolerance, and repair access.
- **Motor-control interface work:** developed practical experience with communication packet design, Arduino/Teensy-side control-board programming, command scaling, calibration, and speed/position/current feedback checks before treating robot motion as valid data.
- **Learning-based locomotion experiments:** ran and documented Isaac Lab / Isaac Sim training using DWAQ/PPO and imitation-learning workflows, tuning foot clearance, posture, contact behavior, gait regularity, torque penalties, action smoothness, and standing behavior.
- **Simulation-to-hardware preparation:** checked model assumptions, observation/action mapping, actuator limits, policy export, ROS2 motor/sensor launches, rosbag recording, topic review, and basic safety conditions before deployment tests.
- **Failure analysis:** compared policy outputs, motor logs, foot-height traces, IMU interpretation, TensorBoard records, simulation videos, and observed robot motion to separate policy issues from actuator, sensor, communication, and assembly problems.
- **Design revision from evidence:** converted physical failures into engineering changes, including encoder/magnet-seat tolerance checks, shaft material and coaxiality concerns, loose linkage correction, calibration changes, and serviceability improvements for repeated tests.

Research signal	Hardware-aware robot learning: actuator packaging, reducer behavior, motor feedback, state estimation, and learned policy behavior are treated as one coupled system rather than separate tasks.
Methods in use	Traceable experiment loop: Isaac Lab / Isaac Sim, DWAQ/PPO, imitation learning, reward shaping, TensorBoard comparison, URDF checks, ROS2 launch workflows, rosbag records, topic-level debugging, and video review.
Hardware interface	Actuator-to-policy interface: actuator/reducer packaging, ODrive/Teensy context, motor speed/position/current feedback, communication packet design, calibration, wiring checks, and fault isolation.
Graduate fit	Graduate research readiness: prepared for robotics labs where mechanical design, embedded control, robot learning, and real-world testing must be evaluated together rather than in isolation.

SELECTED ENGINEERING PROJECTS

Self-Built Simple Robot Dog - Team Lead, Mechanical Designer, Controls Developer

Microprocessor Controlled Systems | A+ outcome | ESP32, PCA9685, MATLAB, Arduino, 3D-printed structure

- **Led a small team** to design and integrate a compact eight-servo quadruped platform around serviceability, wiring, battery placement, center-of-gravity position, and repeated repair during gait testing.
- **Connected simulation to hardware:** translated inverse-kinematics and gait reasoning from MATLAB into embedded ESP32/PCA9685 servo control, separating behavior logic, gait timing, calibration, and servo output for easier debugging.
- **Delivered multiple physical behaviors** including walking, standing, sitting, waving, hip-lift, and rocking while confronting the mismatch between simulated motion, servo limits, wiring, and physical body behavior.

Quadcopter Ball-Transfer Drone - Team Lead and Systems Integrator

Practice of Mechanical Engineering | Full marks and A+ outcome | Peer reference point for many teams

- **Led full-system mission design** for a drone that had to fly, carry, transfer, and release balls reliably under final evaluation pressure.
- **Coordinated cross-domain integration** across airframe, payload mechanism, flight controller, SBUS receiver, ESC/motor setup, spare-system planning, and final-risk control; our architecture and testing process became a reference point for many peer teams.
- **Converted failures into decisions** through thrust-to-throttle tests, drift diagnosis, vibration checks, crash recovery, payload trials, signal-chain debugging, and engineering documentation.

Foldable Ball Transfer Mechanism - Team Lead

Machine Design Theory | Mechanism design, packaging constraint, fabrication-aware decision making

- **Led concept selection and mechanism architecture** for a foldable long-distance ball-transfer mechanism under packaging and deployment constraints.
- **Balanced ideal motion with buildable structure** by evaluating folding architecture, linkage geometry, component accessibility, fabrication feasibility, latch behavior, locked-state stiffness, and tolerance sensitivity.
- **Prioritized reliability under real fabrication limits**, including deployment repeatability, assembly order, repair access, and accessible materials over unnecessary complexity.

LEADERSHIP AND TEAM EXECUTION

Engineering Project and Robotics Team Leadership

Hands-on project teams, CKHS Robotics Team, FRC programming/strategy/scouting leadership

- **Led multiple hands-on engineering teams**, including the self-built robot dog, quadcopter ball-transfer drone, and foldable ball-transfer mechanism projects, translating open-ended goals into design choices, task ownership, testing plans, and final deliverables.
- **Coordinated high-pressure system integration** in the drone project, aligning mechanical design, propulsion testing, electronics, signal debugging, spare parts, and final-run risk control; the finished system became a reference point for many peer teams.
- **Served as programming, strategy, and scouting leader** in CKHS Robotics Team / FRC, connecting autonomous logic, driver needs, mechanism limits, match data, and repair speed during regional and championship-level competition environments.
- **Developed evidence-based technical leadership:** built scouting/Excel analysis workflows, clarified failure symptoms, separated mechanical/electrical/software causes, assigned the next test, and kept the team moving when time was limited.

AWARDS AND RECOGNITION

- **2022 FRC Sacramento Regional:** Finalist and Industrial Design Award.
- **2022 FRC New Taipei City x Hon Hai Regional:** Engineering Inspiration Award; qualified for Houston Championship.
- **2022 FRC Carver Championship Division, Houston:** international championship participation.
- **2023 FRC Los Angeles Regional:** Excellence in Engineering Award.
- **2023 FRC San Diego Regional:** Excellence in Engineering Award.
- **2nd ISS Kibo Robot Programming Challenge Taiwan Preliminary:** Merit Award.
- **58th Hsinchu County Science Fair, Biology Division:** Second Place.

TECHNICAL SKILLS

Mechanical Systems	Mechanism design and actuator packaging; cycloidal/planetary reducer concepts; CAD; 3D printing; machining; engineering drawings; tolerance and assembly planning.
Robot Learning and Control	NVIDIA Isaac Lab / Isaac Sim; DWAQ/PPO workflows; imitation learning; reward shaping; observation/action-space tuning; URDF; ROS2; Pinocchio; MATLAB/Simulink; motor calibration.
Embedded and Software	Embedded motor-control stack: Python; C++; Arduino; ESP32; Teensy 4.0; ODrive; PCA9685; RS485; UART; I2C; PWM; Git/GitHub; Linux; serial diagnostics.
Testing and Documentation	Experimental debugging and documentation: thrust testing; vibration/drift diagnosis; motor speed/position/current feedback; TensorBoard comparison; rosbag/topic checks; failure isolation; engineering reports; portfolio documentation.